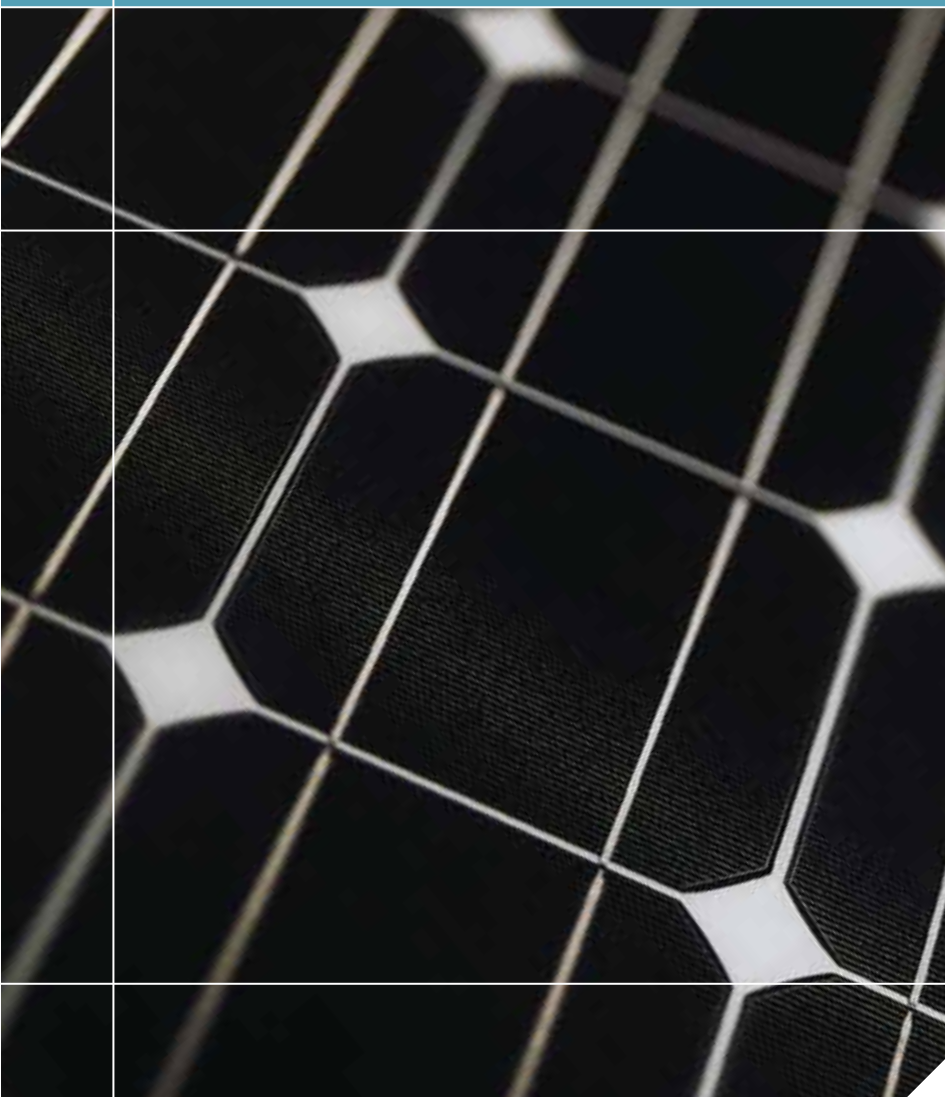
Output classes 150 W - 170 W



solar module aleo s_03

aleo

aleo

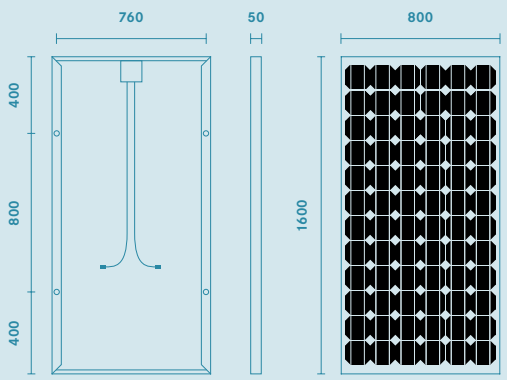
→ solar module aleo s_03 5 inch mono

Specifications	Output class 150 W		Output class 155 W	
Description	aleo S_03 | 150		aleo S_03 | 155	
Data at 1,000 W/m² (STC)[1]				
Rated output	P_{MPP}	150 W	P_{MPP}	155 W
Rated current	I_{MPP}	4.35 A	I_{MPP}	4.46 A
Rated voltage	U_{MPP}	34.5 V	U_{MPP}	34.7 V
Short-circuit current	I_{SC}	4.80 A	I_{SC}	4.91 A
Open-circuit voltage	U_{OC}	42.9 V	U_{OC}	43.1 V
Area-to-power ratio	A_P	8.53 m²/kWp	A_P	8.26 m²/kWp
Efficiency[3]	η(eta)	11.7%	η(eta)	12.1%
Data at 800 W/m² (NOCT)[2]				
Output	P_{MPP}	107 W	P_{MPP}	111 W
Current	I_{MPP}	3.48 A	I_{MPP}	3.56 A
Voltage	U_{MPP}	30.8 V	U_{MPP}	31.1 V
Short-circuit current	I_{SC}	4.03 A	I_{SC}	4.04 A
Open-circuit voltage	U_{OC}	39.1 V	U_{OC}	39.4 V
Efficiency[3]	η(eta)	10.5%	η(eta)	10.8%
Classification range (positive classification)		-0 W/+4.99 W		-0 W/+4.99 W
Measurement accuracy P_{MPP}		-3%/+3%		-3%/+3%
Max. system voltage		1,000 V DC		1,000 V DC
Permissible module load[4]		5,400 Pa		5,400 Pa

[1] Electrical values under standard test conditions (STC): 1,000 W/m²; 25°C; AM 1.5
[2] Electrical values under nominal operating cell temperature (NOCT): 800 W/m²; AM 1.5
[3] For the module surface as a whole (1.28 m²)
[4] In accordance with IEC 61215, 10.16 „Extended load test", installation in accordance with the manual

Datasheet tolerances, except for rated output: +/-10%

Dimensions [mm]



Additional information

Temperature coefficients	α (I_{SC}) β (U_{OC}) γ (P_{MPP})
Certification	IEC/EN 61215, IEC/EN 61730 and protection
Testing organization	VDE
Module dimensions	1600 x 800 x 50 mm
Weight	16 kg

Output class 160 W		Output class 165 W		Output class 170 W	
aleo S_03 | 160		aleo S_03 | 165		aleo S_03 | 170	
P_{MPP}	160 W	P_{MPP}	165 W	P_{MPP}	170 W
I_{MPP}	4.57 A	I_{MPP}	4.69 A	I_{MPP}	4.80 A
U_{MPP}	35.0 V	U_{MPP}	35.2 V	U_{MPP}	35.4 V
I_{SC}	5.03 A	I_{SC}	5.14 A	I_{SC}	5.25 A
U_{OC}	43.3 V	U_{OC}	43.5 V	U_{OC}	43.8 V
A_P	8.00 m²/kWp	A_P	7.76 m²/kWp	A_P	7.53 m²/kWp
η(eta)	12.5%	η(eta)	12.9%	η(eta)	13.3%
P_{MPP}	114 W	P_{MPP}	118 W	P_{MPP}	122 W
I_{MPP}	3.64 A	I_{MPP}	3.72 A	I_{MPP}	3.79 A
U_{MPP}	31.5 V	U_{MPP}	31.8 V	U_{MPP}	32.1 V
I_{SC}	4.05 A	I_{SC}	4.06 A	I_{SC}	4.07 A
U_{OC}	39.8 V	U_{OC}	40.1 V	U_{OC}	40.5 V
η(eta)	11.2%	η(eta)	11.5%	η(eta)	11.9%
	-0 W/+4.99 W		-0 W/+4.99 W		-0 W/+4.99 W
	-3%/+3%		-3%/+3%		-3%/+3%
	1,000 V DC		1,000 V DC		1,000 V DC
	5,400 Pa		5,400 Pa		5,400 Pa

+0.03%/K	**Reduction in efficiency**		< 7%
-0.35%/K	From 1,000 W/m² to 200 W/m²		
-0.48%/K			
class II	**Reverse current load**	I_R	8 A
	NOCT		47°C
	Power guarantee		10 years: 90%, 25 years: 80%

VDE Prüfinstitut

May 2009 | **aleo solar**

solar module aleo s_03

The aleo S_03 solar module is characterized by the state-of-the-art processing of high grade components. 72 monocrystalline silicon cells (5 inch | 125 mm x 125 mm) in each module ensure excellent performance, even with limited solar irradiation. A tight output tolerance of -3%/+3% and purely positive module classification (-0 W/+4.99 W) fulfil the highest standards.

The solar cells are embedded in EVA (ethylene-vinyl acetate), which is resistant to UV radiation. The frame consists of a torsionally rigid, corrosion-resistant aluminium alloy, giving the module stability and allowing it to be mounted in a variety of configurations.

The front panel of the module consists of thermally prestressed solar glass. As well as guaranteeing high light transmittance, the glass also protects the solar cells from external weathering influences such as hail, snow and ice. A polymer backsheet guarantees good insulation and long service life.

The junction box on the back is fitted with bypass diodes to prevent individual solar cells from overheating (hot-spot effect). Several solar modules can easily be connected in series by means of two pre-fitted 1.1 m solar cables with solar plugs.

aleo solar modules are certified according to the European and international IEC/EN 61215 and IEC/EN 61730 standards and fulfil the criteria for protection class II. The power guarantee is at least 90% up to 10 years, and at least 80% up to 25 years, subject to the terms and conditions of the limited warranty.

Please contact your qualified aleo dealer:

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